

# Measuring Error

## Numerical Methods

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# Error Behavior

- Define error behavior using Big-O notation and error terms:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4) \quad \text{where} \quad O(x^n) \leq C |x^n|$$

- Bound differences between errors as parameters change:

$$\begin{array}{llllll} x_1 = 0.50 & n_1 = 3 & \epsilon_1 \leq C |0.5|^3 & \epsilon_1 \leq 0.125C & & \\ x_2 = 0.75 & n_2 = 7 & \epsilon_2 \leq C |0.75|^4 & \epsilon_2 \leq 0.316C & \frac{\epsilon_1}{\epsilon_2} \approx 0.396 & \end{array}$$

- We can calculate how error bounds change without precisely knowing  $\epsilon_1$  or  $\epsilon_2$
- More precise values for  $\epsilon$  require approximating  $C$  in some way:
  - theoretically using features of the function
  - practically by comparing the result to some ground truth

# Approximating Error Analytically

- Consider the Lagrange remainder in a finite Maclaurin series expansion:

$$f(x) = \sum_{n=0}^k \frac{f^{(n)}(0)}{n!} (x)^n + \frac{f^{(k+1)}(z)}{(k+1)!} x^{k+1}$$

$$\frac{f^{(k+1)}(z)}{(k+1)!} x^{k+1} = O(x^{k+1})$$

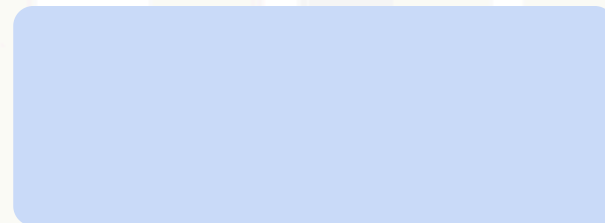
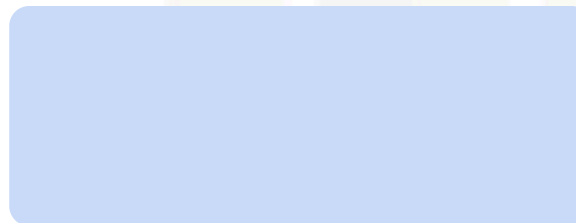
$$\frac{f^{(k+1)}(z)}{(k+1)!} x^{k+1} = C |x|^{k+1}$$

$$C = \frac{f^{(k+1)}(z)}{(k+1)!} \quad \text{where } z \in [0, x]$$

- What is  $\epsilon$  for  $e^x$  using a 5th-order series?

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \epsilon$$

- If  $\frac{d}{dx} e^x = e^x$  then  $f^{(d+1)}(z) = e^z$
- $e^x$  is monotonically increasing, so the largest value for  $z \in [0, x]$  is  $z = x$
- The numerator is **at most**  $e^x$
- The error **in the worst case** is  $\epsilon = \frac{e^x}{6!} x^6$



# Approximating Error Practically

- Absolute error is the magnitude of the difference between true and calculated values:

$$\epsilon_a = |\epsilon| = |t(x) - c(x)|$$

- How do we get the true value  $t(x)$ ?
- Once again, consider  $e^x$  as a 5th-order Maclaurin series:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \epsilon$$

```
1 float exp5(x) {  
2  
3     float ex = 1;  
4     float xp = 1;  
5     float fac = 1;  
6     for(int i = 1; i <= 5; i++){  
7         xp *= x;  
8         fac *= i;  
9         ex += xp / fac;  
10    }  
11    return ex;  
12 }
```

```
// initialize with e^0  
// initialize with x^0  
// initialize 0!
```

`exp5(1.0f) = 2.716667`

NASA has  $e^x$  out to 2M digits:

$e^x \approx 2.718281828459045235 \dots$

$\epsilon_a = |\epsilon| = 0.00161516179 \dots$

analytical result  $\longrightarrow \epsilon \leq 0.00378 \dots$

# Understanding Error

- An error is a deviation of a result from reality
  - a **systematic error** is introduced by **bias** - this can often be correlated
  - a **random error** is a product of **uncertainty** and impossible to correct
  - random errors can often be **controlled** or **bounded**
- Absolute vs. relative error:

$$\epsilon_a = |\epsilon| = |t(x) - c(x)|$$

$$\epsilon_r = \left| \frac{\epsilon_a}{t(x)} \right| = \left| \frac{t(x) - c(x)}{t(x)} \right|$$

- Consider the following true and calculated values:

$$t(x) = 1.5$$

$$c(x) = 1.0$$

$$\epsilon_a = 0.5$$

$$\epsilon_r \approx 0.3 \text{ (30\%)}$$

$$t(x) = 100.5$$

$$c(x) = 100.0$$

$$\epsilon_a = 0.5$$

$$\epsilon_r \approx 0.005 \text{ (0.5\%)}$$

$$t(x) = 0.03$$

$$c(x) = 0.53$$

$$\epsilon_a = 0.5$$

$$\epsilon_r \approx 16.7 \text{ (1670\%)}$$

# Relative Error

- Scientific and engineering applications are less sensitive to small errors in large values
  - $\pm 1$  ton is a lot for a truck but not for a cargo ship
- Relative error can compare values at widely varying scales
- Relative error is often expressed as a **percentage error**:  $\epsilon_r \times 100$
- Most computing systems are designed to minimize relative error
- Undefined when the  $t(x) = 0$ , which can be resolved in a few ways:

**regularization**

$$\epsilon_r = \frac{\epsilon_a}{|t(x)| + \sigma}$$

**specify a value**

$$\epsilon_r = \begin{cases} \frac{\epsilon_a}{|t(x)|} & \text{if } t(x) \neq 0 \\ 1 & \text{otherwise} \end{cases}$$

# Relative Error and Interval Measurements

## Fahrenheit

$$t(x) = 50.0^{\circ}\text{F}$$

$$c(x) = 51.8^{\circ}\text{F}$$

$$\epsilon_a = 1.8^{\circ}\text{F}$$

$$\epsilon_r = 0.036 \text{ (3.6\%)}$$

## Celsius

$$t(x) = 10.0^{\circ}\text{C}$$

$$c(x) = 11.0^{\circ}\text{C}$$

$$\epsilon_a = 1^{\circ}\text{F}$$

$$\epsilon_r = 0.1 \text{ (10\%)}$$

## Kelvin

$$t(x) = 283.0\text{K}$$

$$c(x) = 284.0\text{K}$$

$$\epsilon_a = 1\text{K}$$

$$\epsilon_r = 0.0035 \text{ (0.35\%)}$$

- Relative error requires a **ratio scale** with a **natural zero**
- Ratios have to make sense
  - $2x$ : there is twice some quantity
  - $\frac{1}{3}x$ : there is one third of something
- The zero indicates that there is zero of something